



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—PRELIM PAPER 1—

6 September 2024

Examiner: Mr. A Meintjes
Internal Moderator: Mr. D. Grace
External Moderator: C. Eratosthenes

TIME: 3 hours

MARKS: 150 Marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. Write your name and class in the space provided on the answer book.
 2. Please check that your question paper is complete.
 3. Read the questions carefully and answer ALL the questions in your answer booklet.
 4. It is in your own interest to write legibly and to present your work neatly.
 5. You are provided with a 1 : 50 000 topographic map 3325DC & DD and 3425BA GQEBERHA and an orthophoto 3325 DC 23 GQEBERHA of a part of the mapped area.
 6. You must hand the topographic map and orthophoto map to the invigilator at the end of this examination session. You may only use pencil on these maps. Please erase any pencil markings on the map before handing it back in.
 7. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
 8. Indicate the unit of measurement or compass direction in the final answer of calculations, e.g. 10 km; 2.1cm; west of true north.
 9. You may use a non-programmable calculator.
 10. You may make use of a magnifying glass.
 11. The area demarcated in RED/BLACK on the topographic map represents the area covered by the orthophoto map.
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SECTION A – Climate and Weather

QUESTION 1: Short Questions

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Pressure belts can be defined as ...

- A. A localised area of low pressure such as a cold front.
- B. Areas of differing pressure that extend around the planet going from East to West.
- C. Extended areas of differing pressure found along the coastline.
- D. Areas of differing pressure that extend around the planet going from North to South. (1)

1.1.2. A North facing slope in South Africa would experience ...

- A. Warmer conditions between December and February.
- B. Colder conditions between December and February.
- C. Consistently warmer conditions throughout the year.
- D. Consistently cooler conditions throughout the year. (1)

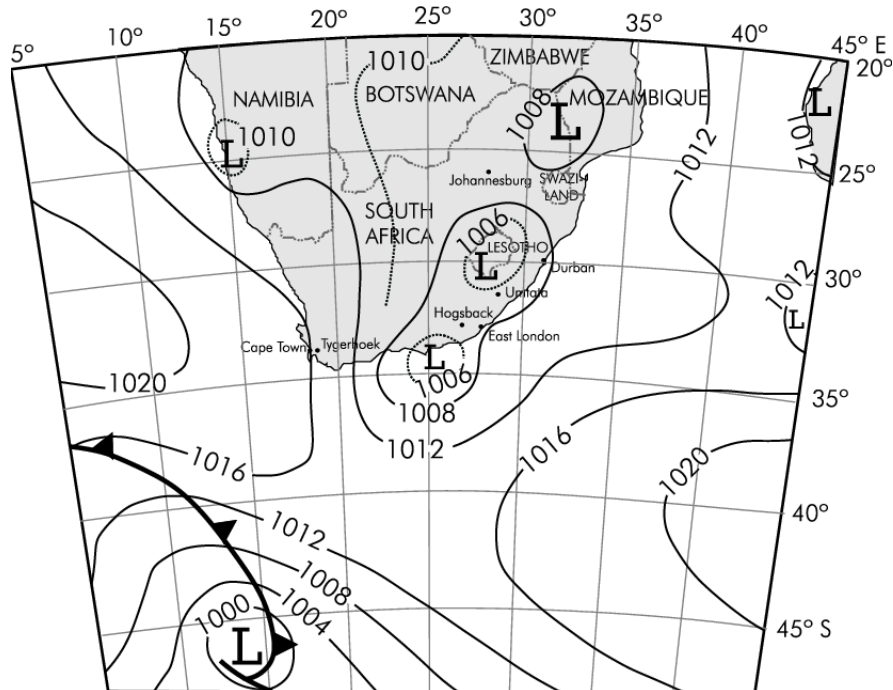
1.1.3. Katabatic winds refer to ...

- A. Winds that blow up valley sides and along the valley floor during the day.
- B. Winds that travel vertically up in cities due to excess heat sources.
- C. Sinking winds resulting from temperature inversions.
- D. Winds that blow down valley sides and along the valley floor during the night. (1)

1.1.4. Anabatic winds refer to ...

- A. Winds that blow up valley sides and along the valley floor during the day.
- B. Winds that travel vertically up in cities due to excess heat sources.
- C. Sinking winds resulting from temperature inversions.
- D. Winds that blow down valley sides and along the valley floor during the night. (1)

Refer to the diagram below showing a typical synoptic weather map and answer the questions that follow:



1.1.5. The season shown above is most likely ...

- A. Summer.
- B. Autumn.
- C. Winter.
- D. Spring.

(1)

1.1.6. The weather phenomenon seen approaching South Africa from the bottom left is a ...

- A. Mid-latitude cyclone.
- B. Hurricane.
- C. Tropical cyclone.
- D. Cut-off low.

(1)

1.1.7. The low pressure cells that are dominating the interior of South Africa have developed because ...

- i. There is an increase in heat this time of the year.
- ii. The Kalahari High Pressure has lifted.
- iii. The interior is hotter than the coastline.
- iv. The cold Benguela current has weakened.

A. i & iii ONLY.

B. None of the above.

C. All of the above.

D. i, ii & iii ONLY.

(1)

1.1.8. Low pressure systems in Summer can lead to ...

A. Colder conditions over a few weeks.

B. Clearer skies.

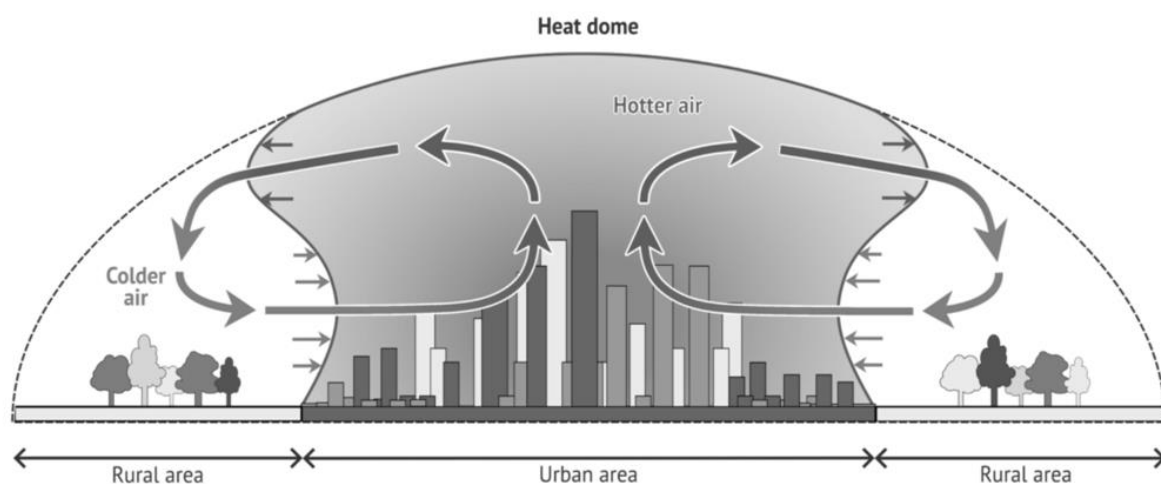
C. Increased flooding risks.

D. Increased temperature over a few days.

(1)

(8 x 1) (8)

1.2. Refer to the diagram below and answer the questions that follow. For each answer select the correct word in brackets. Write only the question and the correct word. E.g. 1.2.8 Jam.



- 1.2.1. Air rising in the urban area would signify an area of (high pressure/low pressure).
- 1.2.2. Rural areas are cooler due to having (increased wind/more vegetation) than urban areas.
- 1.2.3. One reason for the air being warmer over urban areas is (an increase of artificial heating/more exposure to sunlight).
- 1.2.4. Due to the urban heat island effect in summer, urban areas receive (more rainstorms/less rainstorms).
- 1.2.5. The typical type of rain experienced in urban areas as a result of this heat island effect is (orographic/convectional) rain.
- 1.2.6. In winter the urban heat island effect is subdued. This is because there is a lack of (artificial heat sources/sunlight) contributing to the effect.
- 1.2.7. In winter we see the creation of pollution domes. Pollution is trapped over urban cities by (an inversion layer/cold fronts).

(7 x 1) (7)

[15]

QUESTION 2: Tropical Cyclones

2. Refer to the article below and answer the questions that follow:

New forecast reveals when and where hurricane season could get active after Ernesto

By [Mary Gilbert](#), CNN Meteorologist

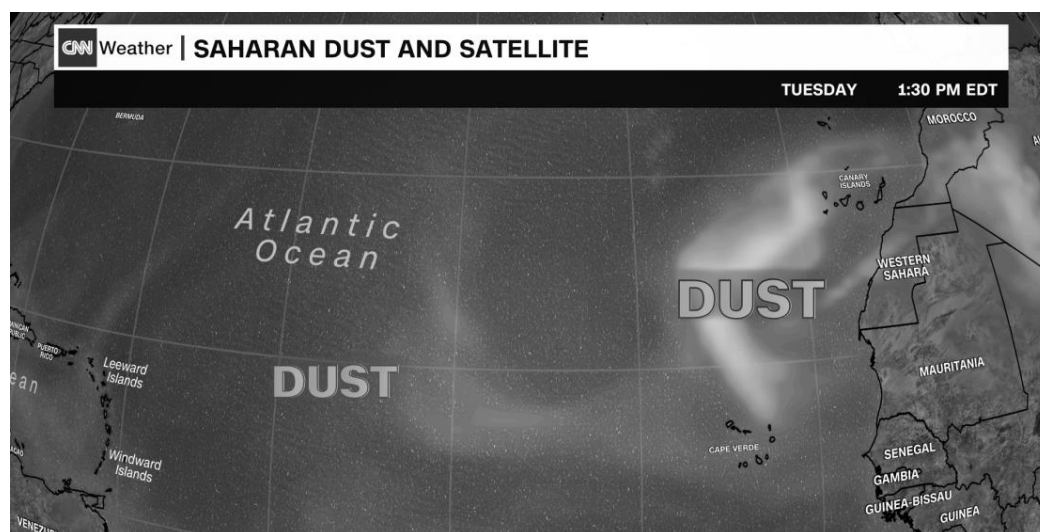
Published 3:19 PM EDT, Tue August 20, 2024

CNN —

Tropical activity in an unusually active hurricane season will take a breather after [Ernesto](#), but forecasters don't believe it will last long.

Atlantic storms will be limited for at least the next week. Dry, dusty air and some disruptive winds in the upper atmosphere are making it difficult for stormy weather to get its act together. A [similar setup](#) kept the Atlantic quiet for a few weeks following record-breaking Hurricane [Beryl](#).

But this brief break in activity isn't likely to last nearly as long, according to forecasts from the Climate Prediction Center. Once the dry, dusty air eases, Atlantic tropical activity could be off to the races as stormy weather pushes off the coast of Africa and out to sea unimpeded.



A plume of dust impacts part of the Atlantic Ocean on Tuesday afternoon.

This isn't surprising because the climatological peak of hurricane season is just a few weeks away.

The Atlantic ocean is also still near record-warm, which could help systems develop and potentially [explode in strength](#) – something becoming more common in a world warming due to fossil fuel pollution.

The season has already been unusually active, producing five named storms – three of which became hurricanes – by mid-August. Ernesto strengthened into the third hurricane of the season [almost a month earlier](#) than normal, according to the NHC.

Beryl also shattered expectations when it became the earliest Category 5 hurricane on record in early July. It was a major hurricane – Category 3 or stronger – almost two months ahead of schedule.

This season has already racked up about half of the activity an entire normal season would produce with plenty of tropical activity to come.

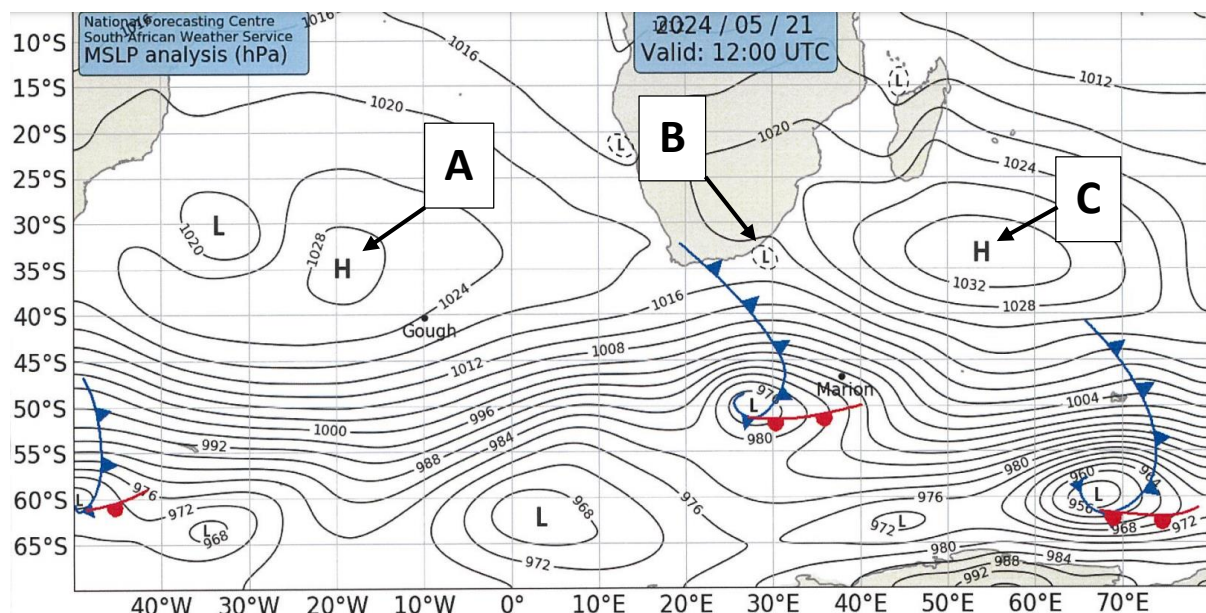
- 2.1. **What** ocean did hurricane Ernesto develop over? (1 x 1) (1)
- 2.2. **How** many hurricanes have developed over the Atlantic this year, including Ernesto? (1 x 2) (2)
- 2.3. According to the article, dry and dusty air is causing a 'break' in hurricane season. **Why** would dry and dusty air cause this 'break'? (1 x 2) (2)
- 2.4. Hurricanes develop over the Atlantic Ocean. **What** is the name we give to these massive storms when they develop over the Indian Ocean? (1 x 1) (1)

- 2.5. When considering hurricane Ernesto, **quote ONE** piece of evidence from the article to support the notion that climate change may be responsible for these storms developing so early. (1 x 1) (1)
- 2.6. In a paragraph of approximately **EIGHT lines**, **discuss** the social impacts of hurricanes and the strategies we can implement to mitigate the damage caused by these phenomena. (4 x 2) (8)

[15]

QUESTION 3: Anti-cyclones

3. Study the synoptic weather map below and answer the questions that follow:

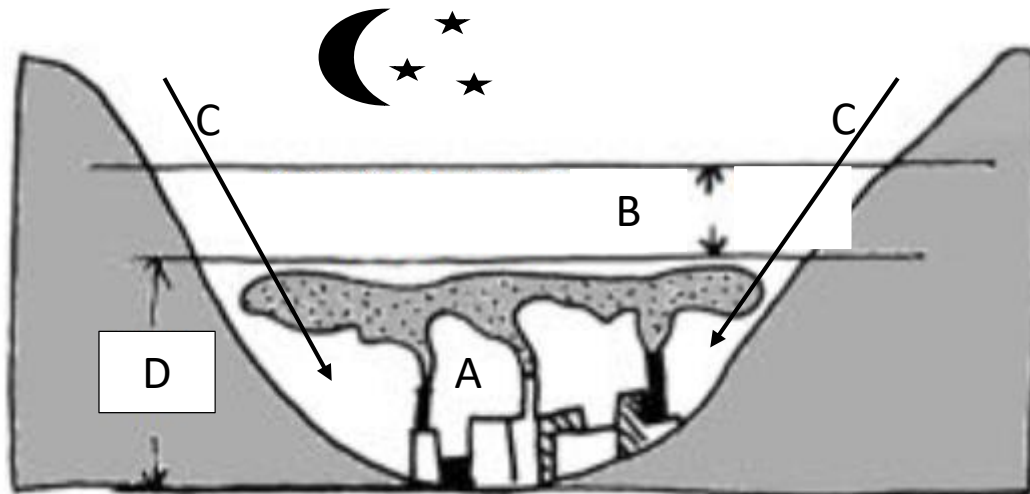


- 3.1. **What** season is this synoptic weather map depicting? (1 x 1) (1)
- 3.2. **Provide** the names for the pressure systems labelled A and C on the map respectively. (2 x 1) (2)
- 3.3. **Draw** a simple cross section showing how the low pressure named at B may lead to the formation of Berg Winds. (4 x 1) (4)
- 3.4. **Discuss** why Berg Winds are a danger to South Africa. Provide **TWO** reasons in your answer (2 x 2) (4)
- 3.5. Not shown on this synoptic weather map is the Kalahari High Pressure cell that dominates the interior of South Africa during this season. **Define** the term *High Pressure*. (1 x 2) (2)

- 3.6. **Describe** the role that the Kalahari High Pressure cell plays in the reduction of rain over the interior during this season. (1 x 2) (2)
- [15]**

QUESTION 4: Valley Climates

4. Study the figure below showing a typical valley climate and answer the questions that follow:



- 4.1. **Name** the wind that flows down a slope at night, labelled C. (1 x 1) (1)
- 4.2. **Explain** how this wind formed. (2 x 2) (4)
- 4.3. **Indicate** if warm or cold air would be found at A and B respectively. (2 x 1) (2)
- 4.4. Fully **explain** how the inversion layer at B would form. (3 x 2) (6)
- 4.5. **List TWO** potential concerns for people settling on the valley floor. (2 x 1) (2)

[15]

TOTAL SECTION A: [60]

SECTION B - Geomorphology

QUESTION 5: Short Questions

5.1. The following descriptions are all features of a river system. State whether each of these is found in the UPPER, MIDDLE or LOWER course of a river.
E.g. 5.1.8. Upper

5.1.1. Erosion is dominant at in the lower course of a river.

5.1.2. In an ideal transverse river profile, erosion and deposition would be equal.

5.1.3. Rapids and waterfalls are a features of the upper river course.

5.1.4. Most headward erosion takes place in the middle course of a river.

5.1.5. Oxbow lakes typically form within deltas.

5.1.6. The Nile travelling through the desert would be an example of an episodic river.

5.1.7. Saltation refers to the percentage of salt in a river.

(7 x 1) (7)

5.2. Match the terms in the table with one of the descriptions. Write only the number of the question and the correct term (5.2.1–5.2.8) in the ANSWER BOOK, for example 5.2.9 Sunshine.

Capacity	Knickpoint	Sinuosity
Valley	Meander	Abrasion
Antecedent drainage	Hydraulic action	Source
Watershed	Confluence	Interfluves
Mouth	Laminar flow	Turbulent flow

5.2.1. The erosion of the river bottom and the riverbank by the material carried in the river.

5.2.2. The water level, or stage, at which a stream, river or lake is at the top of its banks and any further rise would result in water moving into the flood plain.

5.2.3. A ridge or area of land dividing two river valleys within a drainage basin.

5.2.4. A term to describe a location in a river or channel where there is a sharp change in channel slope.

5.2.5. A process whereby the continual colliding of water and other materials on the base of the plunge pool gradually deepens the pool.

5.2.6. The term that refers to the beginning of a river.

5.2.7. An elevated area of land that separates two drainage basins.

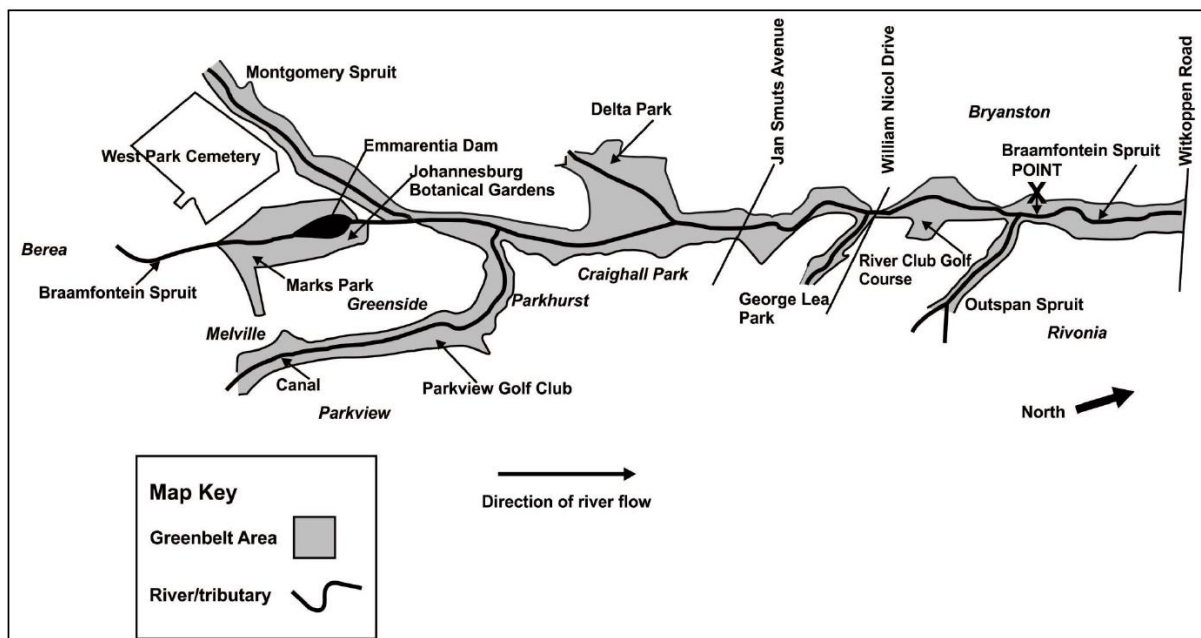
5.2.8. A term used to describe calm water with little turbulence.

(8 x 1) (8)

[15]

QUESTION 6: Drainage basins

6. Refer to the figure below showing a drainage basin and answer the questions that follow:



[Source: <<http://www.amethyst.co.za/Veld/BraamfonteinSpruitBryanston>>]

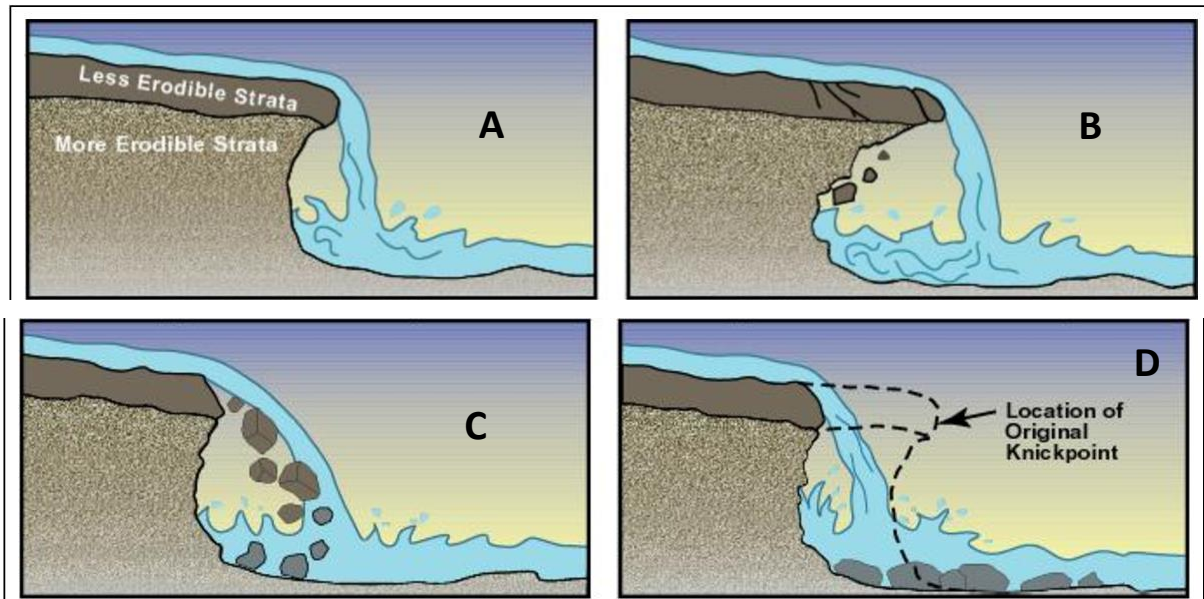
- 6.1. **Name** TWO tributaries of the Braamfontein Spruit? (2 x 1) (2)
- 6.2. Using only the figure above, **determine** the stream order of the Braamfontein Spruit when the river passes Witkoppen Road. (1 x 2) (2)
- 6.3. **Name** ONE temporary base level along the Braamfontein Spruit as seen above. (1 x 1) (1)
- 6.4. With reference to the source above, **suggest THREE** ways in which urban development has impacted upon the natural channel flow of the Braamfontein Spruit. (3 x 2) (6)
- 6.5. Storm water drains in the Braamfontein Spruit catchment are often blocked with rubbish. **Suggest** how this may impact upon:

The **discharge** of the Braamfontein Spruit and
The **health** of the various animals, birds and fish found in this area. (2 x 2) (4)

[15]

QUESTION 7: Fluvial Processes

7. Study the figure below and answer the questions that follow.



- 7.1. **What** feature is being shown in the diagram above? (1 x 1) (1)
- 7.2. **What** is another term for "Less Erodible Strata"? (1 x 1) (1)
- 7.3. **Name** the erosive process occurring at B. (1 x 1) (1)
- 7.4. The feature identified in QUESTION 7.1. is migrating headwards. By referring to the diagram, **explain** how this process works. (3 x 2) (6)
- 7.5. Features like the above often develop as a result of river rejuvenation.
Define the term *River Rejuvenation*. (1 x 2) (2)
- 7.6. **Describe TWO** negative consequences for people and/or animals that live downstream of a river that has been rejuvenated. (2 x 2) (4)

[15]

QUESTION 8: River Management

Study the article below and answer the questions that follow:

SA's rivers and dams can no longer recover from pollution, say water scientists

The National Water Act has failed to meet its objective of improving the quality of water in South Africa's rivers and dams, according to a resource management scientist.

"We release about seven billion litres of sewage every single day into our rivers and dams and the purpose of the National Water Act at its heart is to improve the quality of water resources," said Anthony Turton, a professor at the Centre for Environmental Management at the University of the Free State.

"Many of our rivers and dams have now reached a point of collapse where they can no longer recover so we can state with empirical evidence that the National Water Act has failed; it's failed to meet its objective of improving the quality of water in the rivers and dams."

He was speaking at a recent Tipping Points webinar on "South Africa's Water Whirlpool of Pain", hosted by the Oppenheimer Generations Research and Conservation. There, panellists discussed how the country's water crisis is accelerating beyond its capacity to fix it.

About 80% of the sewage released into waterways comes back into the system either partially treated or, in some cases, untreated. "How long can we continue without seeing human health implications? It's only a question of time," he said, referring to the 30 cholera deaths in Hammanskraal last year.

Sewage does not only contain human waste but also pathogens such as bacteria and viruses, as well as partially metabolised medication. This includes high levels of oestrogen for birth control purposes and partially metabolised antiretroviral medication partially metabolised antibiotics. "It's not inconceivable that multidrug-resistant pathogens will come out of these places.

“We have some evidence now — there’s probably eight to 10 cases that I’m aware of — where people have been in contact with bacteria in the river where they have either injured themselves slightly ... and that has created an open wound in which a certain bacteria has entered the body: flesh-eating bacteria.”

According to Turton, the rate of pollution in rivers and the rate of change in these river systems is “consistent with this notion of a tipping point”.

“The change in the country’s rivers is becoming so rapid that I believe there’s a strong possibility that it will start exceeding our institutional capacity to cope.”

Yet at the same time, public confidence and public trust in the government and the institutions that provide water is under severe pressure. “We need to go out of our way to have communication that is honest and accurate and to have incredible leadership.”

- 8.1. **What** was the objective of the National Water Act? (1 x 1) (1)
- 8.2. According to the article, **why** did the National Water Act fail? (1 x 2) (2)
- 8.3. **What** percentage of sewerage released into rivers re-enters the water system? (1 x 1) (1)
- 8.4. **What** does EIA stand for? (1 x 1) (1)
- 8.5. **Describe ONE** way that EIAs may help improve the quality of our rivers. (1 x 2) (2)
- 8.6. Considering the following quote from the article: “We need to go out of our way to have communication that is honest and accurate and to have incredible leadership.” Write a short paragraph in which you explore what the future may hold for South Africans if we do not improve our current situation, and provide solutions that leadership could consider to improve the situation. (4 x 2) (8)

[15]

TOTAL SECTION B: [60]

QUESTION 9: Calculations

- 9.1. **What** is the name of the stadium located at F on the topographical map? (1 x 1) (1)
- 9.2. **Provide** the coordinates for the Trig Beacon 266 in block D3. (2 x 1) (2)
- 9.3. **Provide** the map code or map reference for the map North of 3325DC. (1 x 1) (1)
- 9.4. Choose the correct option below:

The area covered by the orthophoto map as represented on the topographic map is approximately:

- A) 3885km².
B) 3.885km².
C) 3.885m².
D) 3885m².

(1 x 2) (2)

- 9.5. **Calculate** the current magnetic declination for 2024. Show ALL calculations. Use the following headings as a guide:

The difference in years:

Mean annual change:

Total change:

Magnetic declination for 2024:

(4 x 1) (4)

[10]

QUESTION 10: Interpretation

- 10.1. The Bakensrivier exists the drainage basin at block C5.
- 10.1.1. **What** is the term given to the part of the river where it meets the ocean? (1 x 1) (1)
- 10.1.2. In **what** course would this river be, middle course or low course? (1 x 1) (1)
- 10.1.3. This river system has a medium drainage density. **Discuss TWO** factors that would influence this system's drainage density. (2 x 2) (4)

- 10.2. Refer to the vast amount of industrialisation in this area (H).
- 10.2.1. **How** may this industrialisation lead to acid rain in Gqeberha in Winter? (1 x 2) (2)
- 10.2.2. Pollution domes are a major concern for this area. **Draw** a simple diagram showing the formation of a pollution dome. (2 x 1) (2)
- 10.2.3. **Suggest ONE** way that the people of Gqeberha can combat the effects of pollution domes. (1 x 2) (2)
- [12]**

QUESTION 11: GIS

Refer to the orthophoto and answer the questions that follow.

- 11.1. **Is** the orthophoto an example of raster or vector data? (1 x 1) (1)
- 11.2. **What** is the feature found between points 6 and 7? (1 x 1) (1)
- 11.3. Buffering was clearly used before building this feature. **Provide** evidence from the map to justify this statement. (1 x 1) (1)
- 11.4. **Why** would we need to use buffering before building the feature identified in QUESTION 11.2? List TWO reason. (2 x 1) (2)
- 11.5. Wind patterns are important for the safe functioning of the feature identified in QUESTION 11.2. **Would** GIS have been used to study the wind patterns before construction? (1 x 1) (1)
- 11.6. **How** would the GIS wind data have been used in this construction? (1 x 2) (2)
- [8]**

TOTAL SECTION C: [30]

GRAND TOTAL: 150 Marks